Probability Theory

STAT 547C & Stochastic Processes Reading Group

2025/2026

Foreword

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1 Measure Theoretic Foundations

1.1 Probability Spaces

Definition 1.1. A **probability space** is a measure space $(\Omega, \mathcal{H}, \mathbb{P})$ such that the total mass is unit, i.e., $\mathbb{P}(\Omega) = 1$. Elements $\omega \in \Omega$ are called outcomes, and sets $A \in \mathcal{H}$ are called events.

Definition 1.2. Let $(\Omega, \mathcal{H}, \mathbb{P})$ be a probability space and $(\mathcal{X}, \mathcal{M})$ be a measurable space. A **random variable** X is a measurable function

$$X : \Omega \to \mathcal{X}.$$

That is, for every $A \in \mathcal{M}$, the pre-image $X^{-1}(A) := \{\omega \in \Omega : X(\omega) \in A\}$ is an element of $\mathcal{H}$. When $(\mathcal{X}, \mathcal{M}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, we simply call X a real-valued random variable.

Definition 1.3. Let $(\mathcal{Y}, \mathcal{N})$ and $(\mathcal{X}, \mathcal{M})$ be measurable spaces. Let $h : \mathcal{Y} \to \mathcal{X}$ be a measurable function and ν be a measure on $(\mathcal{Y}, \mathcal{N})$. The **pushforward measure** of ν by h, denoted by $\nu \circ h^{-1}$ or $h_{\#}\nu$, is the measure on $(\mathcal{X}, \mathcal{M})$ defined by:

$$\left(\nu \circ h^{-1}\right)(A) := \nu(h^{-1}(A)) \quad \text{for all } A \in \mathcal{M},$$

where $h^{-1}(A)$ is the pre-image of A under h.

Proposition 1.4. *(Change of Variables, or Transformation Theorem)* Let $(\mathcal{Y}, \mathcal{N})$ and $(\mathcal{X}, \mathcal{M})$ be measurable spaces. Let $h : \mathcal{Y} \to \mathcal{X}$ be a measurable function and ν be a measure on $(\mathcal{Y}, \mathcal{N})$. Then

$$\int f \, \mathrm{d}\left(\nu \circ h^{-1}\right) = \int f \circ h \, \mathrm{d}\nu$$

for every positive $\mathcal{M}$-measurable function f

Proof. First consider the case where $f = \chi_A$ for some $A \in \mathcal{M}$. Then:

$$\int \chi_A \, \mathrm{d}(\nu \circ h^{-1}) = (\nu \circ h^{-1})(A) = \nu(h^{-1}(A)).$$

On the other hand

$$\int \chi_A \circ h \, \mathrm{d}\nu = \int \chi_{h^{-1}(A)} \, \mathrm{d}\nu = \nu(h^{-1}(A)).$$

Thus, the equality holds for simple functions by linearity. Now let f be any positive $\mathcal{M}$-measurable function. There exists an increasing sequence of non-negative simple functions $s_n \to f$. By the Monotone Convergence Theorem

$$\int f \, \mathrm{d}(\nu \circ h^{-1}) = \lim_{n\to\infty} \int s_n \, \mathrm{d}(\nu \circ h^{-1}) = \lim_{n\to\infty} \int s_n \circ h \, \mathrm{d}\nu.$$

Since $s_n \uparrow f$ implies $s_n \circ h \uparrow f \circ h$, applying MCT again to the right side yields:

$$\lim_{n\to\infty} \int s_n \circ h \, \mathrm{d}\nu = \int f \circ h \, \mathrm{d}\nu.$$

$\square$

Definition 1.5. Let $(\Omega, \mathcal{H}, \mathbb{P})$ be a probability space. If $X : \Omega \to \mathcal{X}$ is a random variable, the pushforward measure $\mathbb{P} \circ X^{-1}$ is called the **distribution** (or **law**) of X. We typically write:

$$\mathbb{P}(X \in A) := (\mathbb{P} \circ X^{-1})(A).$$

We use the notation $X \sim \mu$ to indicate that the distribution of X is the measure μ on $(\mathcal{X}, \mathcal{M})$.

Proposition 1.6. To specify the distribution of a random variable on $(\mathbb{R}, \mathcal{B}(\mathbb{R})$ it suffices to specify the *cumulative distribution function* (or CDF) which is a function $F : \mathbb{R} \to [0, 1]$ such that

$$F(x) = \mathbb{P}(X \in (-\infty, x]) = \mathbb{P}(X \leq x).$$

Proof. We first show that F is non-decreasing and right-continuous.

- Non-decreasing: Let $x < y$. Then $(-\infty, x] \subset (-\infty, y]$. By the monotonicity of measures:

$$F(x) = \mathbb{P}(X \in (-\infty, x]) \leq \mathbb{P}(X \in (-\infty, y]) = F(y).$$

- Right-continuous: Let $x \in \mathbb{R}$ and let $(x_n)_{n \in \mathbb{N}}$ be a decreasing sequence such that $x_n \to x$. Let $A_n = (-\infty, x_n]$. These sets are nested, their intersection is $\bigcap_{n=1}^{\infty} A_n = (-\infty, x]$, and $\mathbb{P}(X \in A_1) = \mathbb{P}(X \leq x_1) = F(x_1) \leq 1 < \infty$. So by continuity from above:

$$\lim_{n \to \infty} F(x_n) = \lim_{n \to \infty} \mathbb{P}(X \in A_n) = \mathbb{P}\left(X \in \bigcap_{n=1}^{\infty} A_n\right) = \mathbb{P}(X \in (-\infty, x]) = F(x).$$

Let $\mathcal{A}$ be the algebra of finite unions of half-open intervals $(a, b]$ in $\mathbb{R}$. Namely, let

$$\mathcal{A} = \{\emptyset\} \cup \left\{ \bigcup_{j=1}^{n} (a_j, b_j] : n \in \mathcal{N}, -\infty \leq a_1 < b_1 < a_2 < \cdots < b_n \leq \infty \right\}.$$

And let $\mu_0(\emptyset) = 0$, $\mu_0\left(\bigcup_{j=1}^{n} (a_j, b_j]\right) = \sum_{j=1}^{n} (F(b_j) - F(a_j))$. One can show that $\mathcal{A}$ is an algebra and that, because F is right-continuous, μ_0 is a pre-measure. By the Carathéodory Extension Theorem, this pre-measure extends to a measure on $\mathcal{M}(\mathcal{A}) = \mathcal{B}(\mathbb{R})$. And since $\mu_0(\mathbb{R}) = \mu(\mathbb{R}) = 1$ is σ-finite, this extension is unique. $\square$

Definition 1.7. Let X be a real-valued random variable. The **expectation** of X is

$$\mathbb{E}X = \int X(\omega) \, \mathrm{d}\mathbb{P}(\omega)$$

Proposition 1.8. Let X be a random variable in $(\mathcal{X}, \mathcal{M})$. Then μ is the distribution of X if and only if

$$\mathbb{E}\left[f \circ X\right] = \int f \, \mathrm{d}\mu$$

for all positive $\mathcal{M}$-measurable f.

Proof. Suppose $\mu = \mathbb{P} \circ X^{-1}$. Then for any positive $\mathcal{M}$-measurable f change of variables yields

$$\int f \, \mathrm{d}\mu = \int f \, \mathrm{d}\big(\mathbb{P} \circ X^{-1}\big) = \int f \circ X \, \mathrm{d}\mathbb{P} = \mathbb{E}\left[f \circ X\right].$$

Conversely, suppose that $\mathbb{E}[f \circ X] = \int f \, \mathrm{d}\mu$ for all positive $\mathcal{M}$-measurable f. Take $f = \chi_A$ for any $A \in \mathcal{M}$. Then

$$\mu(A) = \int \chi_A \, \mathrm{d}\mu = \mathbb{E}[\chi_A \circ X] = \int \chi_A \circ X \, \mathrm{d}\mathbb{P} = \int \chi_A \, \mathrm{d}\big(\mathbb{P} \circ X^{-1}\big) = \mathbb{P}(X \in A)$$

as desired. □

Definition 1.9. Certain expectations have special names and are of particular interest. Let X be a real-valued random variable, $n \in \mathbb{N}$, and r a real-valued constant.

- $\mathbb{E}X^n$ is the **n-th moment of** X and $\mathbb{E}[(X - \mathbb{E}X)^n]$ is the **n-th central moment of** X
- $\mathbb{E}\left[(X - \mathbb{E}X)^2\right]$ is the **variance of** X
- When $X, r \geq 0$, $\mathbb{E}[\exp(-rX)]$ is the **Laplace transform of** X
- $\mathbb{E}[\exp(rX)]$ is the **Moment-generating function of** X
- $\mathbb{E}[\exp(irX)]$ is the **Fourier transform** (sometimes **characteristic function**) **of** X .

Proposition 1.10. Let X be a positive real-valued random variable, and let $r \geq 0$. The Laplace transform $h : [0, \infty] \to [0, \infty]$ given by

$$h(r) = \mathbb{E}[\exp(-rX)]$$

is continuous and decreasing on $(0, \infty)$ and is continuous at 0 if and only if $\mathbb{P}(X < \infty) = 1$.

Proof. Let $X \geq 0$ and $h(r) = \mathbb{E}[e^{-rX}]$.

- Decreasing: Let $0 \leq r_1 < r_2$. Since $X \geq 0$, we have $-r_2 X \leq -r_1 X$. Since the exponential function is strictly increasing, $e^{-r_2 X} \leq e^{-r_1 X}$ almost surely. By the monotonicity of the integral:
$$h(r_2) = \mathbb{E}[e^{-r_2 X}] \leq \mathbb{E}[e^{-r_1 X}] = h(r_1).$$
- Continuity on $(0, \infty)$: Let $r \in (0, \infty)$ and let $(r_n)_{n \in \mathbb{N}}$ be a sequence such that $r_n \to r$. Define $f_n(\omega) = \exp(-r_n X(\omega))$. Since the function $r \mapsto e^{-rx}$ is continuous for any x, $f_n \to \exp(-rX)$ pointwise. Since $r_n, X \geq 0$, we have $|f_n| \leq 1$ for all n. Because $\mathbb{P}$ is a probability measure, the constant function 1 is integrable. Thus, by the Dominated Convergence Theorem:
$$\lim_{n \to \infty} h(r_n) = \lim_{n \to \infty} \mathbb{E}[f_n] = \mathbb{E}[\lim_{n \to \infty} f_n] = \mathbb{E}[e^{-rX}] = h(r).$$
- Continuity at 0: Note that $h(0) = \mathbb{E}[e^0] = \mathbb{E}[1] = 1$. Let $r_n \downarrow 0$. We examine the pointwise limit of $f_n = e^{-r_n X}$:
  - If $X(\omega) < \infty$, then $\lim_{n \to \infty} e^{-r_n X(\omega)} = e^0 = 1$.
  - If $X(\omega) = \infty$, then $e^{-r_n X(\omega)} = 0$ for all n (since $r_n > 0$), so the limit is 0.

  Thus, $e^{-r_n X} \to \chi_{\{X < \infty\}}$ pointwise. Again, by DCT:
$$\lim_{r \to 0^+} h(r) = \mathbb{E}[\chi_{\{X < \infty\}}] = \mathbb{P}(X < \infty).$$
  Hence For h to be continuous at 0, we require $\mathbb{P}(X < \infty) = h(0) = 1$.

□

Definition 1.11. A function $\varphi : (a, b) \to \mathbb{R}$ is **convex** if for all $x, y \in (a, b)$ and $\lambda \in [0, 1]$,

$$\varphi(\lambda x + (1 - \lambda)y) \leq \lambda\varphi(x) + (1 - \lambda)\varphi(y).$$

Geometrically, this means the chord between any two points on the graph lies above the graph.

Theorem 1.12. *(Jensen's Inequality)* Let $(\Omega, \mathcal{H}, \mathbb{P})$ be a probability space and X an integrable real-valued random variable. If $\varphi : \mathbb{R} \to \mathbb{R}$ is convex, then

$$\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)].$$

Proof. One can show that if φ is convex, then for every x_0, there exists $L(x) = ax + b$ such that $L(x_0) = \varphi(x_0)$ and $L(x) \leq \varphi(x)$ for all x. Let $x_0 = \mathbb{E}[X]$. Then there exist $a, b \in \mathbb{R}$ such that $\varphi(\mathbb{E}[X]) = a\mathbb{E}[X] + b$ and $L(X) \leq \varphi(X)$ almost surely. Then by the linearity and monotonicity of expectation:

$$\mathbb{E}[\varphi(X)] \geq \mathbb{E}[aX + b] = a\mathbb{E}[X] + b = \varphi(\mathbb{E}[X]).$$

□

1.2 Probability Kernels

Definition 1.13. Let $(\mathcal{X}, \mathcal{M})$ and $(\mathcal{Y}, \mathcal{N})$ be measurable spaces. Let $\overline{\mathbb{R}}_+ = \{z \in \mathbb{R} : z \geq 0\} \cup \{+\infty\}$. A **kernel from $\mathcal{X}$ into $\mathcal{Y}$** is a function $K : \mathcal{X} \times \mathcal{Y} \to \overline{\mathbb{R}}_+$ such that

(i) For every $E \in \mathcal{N}$, the map $x \mapsto K(x, E)$ is $\mathcal{M}$-measurable

(ii) For every $x \in \mathcal{X}$, the map $B \mapsto K(x, E)$ is a measure on $(\mathcal{Y}, \mathcal{N})$

Definition 1.14. The finiteness properties of measures naturally extend to kernels, categorized by whether the constraints hold pointwise for each $x \in \mathcal{X}$ or uniformly across the domain.

(i) K is a **probability kernel** if $K(x, \mathcal{Y}) = 1$ for all $x \in \mathcal{X}$.

(ii) **Finiteness and Boundedness:**

- K is **finite** if $K(x, \mathcal{Y}) < \infty$ for all $x \in \mathcal{X}$.
- K is **bounded** if it is uniformly finite, i.e., $\sup_{x \in \mathcal{X}} K(x, \mathcal{Y}) < \infty$.

(iii) **σ-Finiteness and σ-Boundedness:**

- K is **σ-finite** if for every $x \in \mathcal{X}$, there exists a countable partition $\mathcal{Y} = \bigcup_{n \in \mathbb{N}} E_n$ such that $K(x, E_n) < \infty$ for all $n \in \mathbb{N}$.
- K is **σ-bounded** if there exists a countable partition $\mathcal{Y} = \bigcup_{n \in \mathbb{N}} E_n$ that is independent of x, such that $\sup_{x \in \mathcal{X}} K(x, E_n) < \infty$ for all $n \in \mathbb{N}$.

(iv) **Σ-Finiteness and Σ-Boundedness:**

- K is **Σ-finite** if it can be decomposed into a sum of finite kernels, $K = \sum_{n \in \mathbb{N}} K_n$.
- K is **Σ-bounded** if it can be decomposed into a sum of bounded kernels, $K = \sum_{n \in \mathbb{N}} K_n$.

Definition 1.15. A kernel from $(\mathcal{X}, \mathcal{M})$ into itself, is simply called a **kernel on** $(\mathcal{X}, \mathcal{M})$.

- K is a **Markov kernel** if $K(x, \mathcal{X}) = 1$ for all $x \in \mathcal{X}$. In this case, $K(x, \cdot)$ is a probability measure for every x.
- K is a **sub-Markov kernel** if $K(x, \mathcal{X}) \leq 1$ for all $x \in \mathcal{X}$.

Proposition 1.16. Let K be a kernel from $(\mathcal{X}, \mathcal{M})$ into $(\mathcal{Y}, \mathcal{N})$, and let L be a kernel from $(\mathcal{Y}, \mathcal{N})$ into $(\mathcal{Z}, \mathcal{H})$. The **composition** KL, defined for $x \in \mathcal{X}$ and $B \in \mathcal{H}$ by

$$KL(x, B) := \int_{\mathcal{Y}} L(y, B) K(x, \mathrm{d}y),$$

is a kernel from $(\mathcal{X}, \mathcal{M})$ into $(\mathcal{Z}, \mathcal{H})$.

Proof. To show that KL is a kernel, we must verify two conditions: that $KL(x, \cdot)$ is a measure for each fixed x, and that $KL(\cdot, B)$ is a measurable function for each fixed B.

- Fix $x \in \mathcal{X}$.
  - Since $L(y, \emptyset) = 0$ for all $y \in \mathcal{Y}$, we have $KL(x, \emptyset) = \int_{\mathcal{Y}} 0 \, K(x, \mathrm{d}y) = 0$.
  - Let $\{B_n\}_{n \in \mathbb{N}}$ be a sequence of disjoint sets in $\mathcal{H}$, and let $B = \bigcup_{n=1}^{\infty} B_n$. Since $L(y, \cdot)$ is a measure for each y, we have $L(y, B) = \sum_{n=1}^{\infty} L(y, B_n)$. Then:

    $$KL(x, B) = \int_{\mathcal{Y}} \sum_{n=1}^{\infty} L(y, B_n) K(x, \mathrm{d}y).$$

    By the Monotone Convergence Theorem, we can interchange the sum and the integral:

    $$KL(x, B) = \sum_{n=1}^{\infty} \int_{\mathcal{Y}} L(y, B_n) K(x, \mathrm{d}y) = \sum_{n=1}^{\infty} KL(x, B_n).$$

    Thus, $KL(x, \cdot)$ is a measure on $(\mathcal{Z}, \mathcal{H})$.
- Fix $B \in \mathcal{H}$. We must show that $g(x) = KL(x, B)$ is $\mathcal{M}$-measurable. By definition of L being a kernel, the function $f(y) = L(y, B)$ is $\mathcal{N}$-measurable and non-negative.
  - If $f = \chi_A$ for some $A \in \mathcal{N}$, then $g(x) = \int \chi_A(y) K(x, \mathrm{d}y) = K(x, A)$, which is $\mathcal{M}$-measurable by the definition of K.
  - By linearity, the property holds for all non-negative simple functions $s = \sum c_i \chi_{A_i}$.
  - For a general non-negative measurable f, there exists a sequence of simple functions $s_n \uparrow f$. By the Monotone Convergence Theorem:

    $$g(x) = \int f(y) K(x, \mathrm{d}y) = \lim_{n \to \infty} \int s_n(y) K(x, \mathrm{d}y).$$

    Each $g_n(x) = \int s_n(y) K(x, \mathrm{d}y)$ is measurable. Since the pointwise limit of measurable functions is measurable, $g(x)$ is $\mathcal{M}$-measurable.

$\square$

Proposition 1.17. Let K be a kernel from $(\mathcal{X}, \mathcal{M})$ into $(\mathcal{Y}, \mathcal{N})$, μ be a measure on $(\mathcal{X}, \mathcal{M})$, and f be a non-negative $\mathcal{N}$-measurable function.

(i) The mapping $\mu K : \mathcal{N} \to [0, \infty]$ defined by

$$(\mu K)(B) := \int_{\mathcal{X}} K(x, B) \mu(\mathrm{d}x)$$

is a measure on $(\mathcal{Y}, \mathcal{N})$.

(ii) The mapping $Kf : \mathcal{X} \to [0, \infty]$ defined by

$$(Kf)(x) := \int_{\mathcal{Y}} f(y) K(x, \mathrm{d}y)$$

is a non-negative $\mathcal{M}$-measurable function.

(iii) The following identity holds:

$$\int_{\mathcal{Y}} f \,\mathrm{d}(\mu K) = \int_{\mathcal{X}} (Kf) \,\mathrm{d}\mu = \int_{\mathcal{X}} \int_{\mathcal{Y}} f(y) K(x, \mathrm{d}y) \mu(\mathrm{d}x).$$

Note 1.18. In the language of probability, if μ is the distribution of X and K is the conditional distribution of Y given X, then μK is the *marginal distribution* of Y, $Kf(x)$ is the *conditional expectation* of $f(Y)$ given $X = x$, an $\int f \,\mathrm{d}(\mu K)$ is the *marginal expectation* of $f(Y)$. Part (iii) is the general version of the *Law of Total Expectation*: $\mathbb{E}[f(Y)] = \mathbb{E}[\mathbb{E}[f(Y)|X]]$.

Proof.

(i) μK is a measure:

- $(\mu K)(\emptyset) = \int_{\mathcal{X}} K(x, \emptyset) \mu(\mathrm{d}x) = \int_{\mathcal{X}} 0 \,\mu(\mathrm{d}x) = 0$.
- Let $\{B_n\}_{n \in \mathbb{N}}$ be a sequence of disjoint sets in $\mathcal{N}$. Since $K(x, \cdot)$ is a measure for each x:

$$(\mu K)\left(\bigcup_{n=1}^{\infty} B_n\right) = \int_{\mathcal{X}} K\left(x, \bigcup B_n\right) \mu(\mathrm{d}x) = \int_{\mathcal{X}} \sum_{n=1}^{\infty} K(x, B_n) \mu(\mathrm{d}x).$$

By the Monotone Convergence Theorem, we interchange the sum and integral:

$$\sum_{n=1}^{\infty} \int_{\mathcal{X}} K(x, B_n) \mu(\mathrm{d}x) = \sum_{n=1}^{\infty} (\mu K)(B_n).$$

Thus, μK is a measure on $(\mathcal{Y}, \mathcal{N})$.

(ii) Kf is $\mathcal{M}$-measurable: This is a special case of prop:kernel-measurability where f does not depend on x.

- If $f = \chi_B$, then $(Kf)(x) = K(x, B)$, which is $\mathcal{M}$-measurable by definition.
- Extension to simple functions follows by linearity, and to positive measurable functions by the Monotone Convergence Theorem.

(iii) We prove $\int f \,\mathrm{d}(\mu K) = \int (Kf) \,\mathrm{d}\mu$ via functional induction on f:

- If $f = \chi_B$, then $\int \chi_B \,\mathrm{d}(\mu K) = (\mu K)(B)$. By definition, $(\mu K)(B) = \int_{\mathcal{X}} K(x, B) \mu(\mathrm{d}x)$. Since $K(x, B) = (K\chi_B)(x)$, the identity holds.
- By linearity, it holds for all non-negative simple functions.
- For positive $\mathcal{N}$-measurable f, let $s_n \uparrow f$. By the Monotone Convergence Theorem applied twice (once for the measure μK and once for the function sequence $Ks_n \uparrow Kf$):

$$\int f \,\mathrm{d}(\mu K) = \lim_{n \to \infty} \int s_n \,\mathrm{d}(\mu K) = \lim_{n \to \infty} \int (Ks_n) \,\mathrm{d}\mu = \int (Kf) \,\mathrm{d}\mu.$$

□

Example 1.19. *(Pushforward as a Kernel)* Let $h : \mathcal{X} \to \mathcal{Y}$ be a measurable function. We can represent the pushforward measure $h_{\#}\mu$ using the kernel $K(x, B) = \delta_{h(x)}(B)$, where δ is the Dirac measure. Then:

$$(\mu K)(B) = \int_{\mathcal{X}} \delta_{h(x)}(B)\mu(\mathrm{d}x) = \int_{\mathcal{X}} \chi_{h^{-1}(B)}(x)\mu(\mathrm{d}x) = \mu(h^{-1}(B)).$$

Thus, $\mu K = h_{\#}\mu$. This shows that kernels generalize the notion of measurable transformations.

Proposition 1.20. Let K be a Σ-finite kernel from $(\mathcal{X}, \mathcal{M})$ into $(\mathcal{Y}, \mathcal{N})$ and let μ be a measure on $(\mathcal{X}, \mathcal{M})$. There exists a unique measure π on the product space $(\mathcal{X} \times \mathcal{Y}, \mathcal{M} \otimes \mathcal{N})$ such that for all $A \in \mathcal{M}$ and $B \in \mathcal{N}$:

$$\pi(A \times B) = \int_A K(x, B)\mu(\mathrm{d}x).$$

For any non-negative $(\mathcal{M} \otimes \mathcal{N})$-measurable function f, the measure π satisfies:

$$\int_{\mathcal{X}\times\mathcal{Y}} f \,\mathrm{d}\pi = \int_{\mathcal{X}} \left(\int_{\mathcal{Y}} f(x, y)K(x, \mathrm{d}y) \right) \mu(\mathrm{d}x).$$

Furthermore, if μ is σ-finite and K is σ-bounded, then π is σ-finite.

Proof. We construct the measure π by defining it on an algebra and extending it.

1. **Definition on the Algebra:** Let $\mathcal{A}$ be the algebra of finite unions of measurable rectangles. For a rectangle $A \times B$, define

   $$\pi_0(A \times B) = \int_A K(x, B)\mu(\mathrm{d}x).$$

   This set function is well-defined and additive on $\mathcal{A}$. To show π_0 is a pre-measure, we check countable additivity. Let $C = \bigcup C_n$ with $C, C_n \in \mathcal{A}$ disjoint. By applying the Monotone Convergence Theorem to the integral of the kernel, one can show $\pi_0(C) = \sum \pi_0(C_n)$.

2. **Existence and Uniqueness:** By the Carathéodory Extension Theorem, π_0 extends to a measure π on $\sigma(\mathcal{A}) = \mathcal{M} \otimes \mathcal{N}$. If μ is σ-finite and K is σ-bounded, there exist partitions $\{A_i\} \subset \mathcal{M}$ and $\{B_j\} \subset \mathcal{N}$ with $\mu(A_i) < \infty$ and $\sup_x K(x, B_j) < \infty$. The rectangles $A_i \times B_j$ form a countable partition of $\mathcal{X} \times \mathcal{Y}$ with $\pi(A_i \times B_j) < \infty$, ensuring π is σ-finite and the extension is unique.

3. **Integral Identity:** We show $\int f \,\mathrm{d}\pi = \int (Tf) \,\mathrm{d}\mu$ (where Tf is the operator from prop:kernel-measure-function).

   - If $f = \chi_{A\times B}$, then $\int \chi_{A\times B} \,\mathrm{d}\pi = \pi(A \times B)$. By definition:

     $$\pi(A \times B) = \int_{\mathcal{X}} \chi_A(x)K(x, B)\mu(\mathrm{d}x) = \int_{\mathcal{X}} \left(\int_{\mathcal{Y}} \chi_{A\times B}(x, y)K(x, \mathrm{d}y) \right) \mu(\mathrm{d}x).$$

   - Let $\mathcal{C} = \{C \in \mathcal{M} \otimes \mathcal{N} : \text{the identity holds for } f = \chi_C\}$. We have shown rectangles are in $\mathcal{C}$. Using the linearity of the integral and the Monotone Class Lemma, $\mathcal{C} = \mathcal{M} \otimes \mathcal{N}$.

- The identity extends to non-negative simple functions by linearity and to all positive $(\mathcal{M}\otimes\mathcal{N})$-measurable f by the Monotone Convergence Theorem, noting that prop:kernel-measurability guarantees the measurability of the inner integral.

□

Proposition 1.21. Let K be a Σ-finite kernel from $(\mathcal{X}, \mathcal{M})$ into $(\mathcal{Y}, \mathcal{N})$. For every non-negative $(\mathcal{M} \otimes \mathcal{N})$-measurable function $f : \mathcal{X} \times \mathcal{Y} \to [0, \infty]$, the function $Tf : \mathcal{X} \to [0, \infty]$ given by

$$Tf(x) = \int_{\mathcal{Y}} f(x, y)K(x, \mathrm{d}y)$$

is $\mathcal{M}$-measurable.

Proof. Assume first that K is a finite kernel. We proceed by showing that the property of "measurability under the integral" extends from a simple algebra to the full product σ-algebra.

- Let $\mathcal{A}$ be the algebra consisting of finite unions of measurable rectangles $A \times B$ (where $A \in \mathcal{M}, B \in \mathcal{N}$). The product σ-algebra $\mathcal{M} \otimes \mathcal{N}$ is the σ-algebra generated by this algebra.
- Let $C \in \mathcal{A}$. Then χ_C can be written as a finite sum of indicators of disjoint rectangles $\sum_{i=1}^{n} \chi_{A_i \times B_i}$. For each rectangle:

  $$T\chi_{A_i \times B_i}(x) = \int_{\mathcal{Y}} \chi_{A_i}(x)\chi_{B_i}(y)K(x, \mathrm{d}y) = \chi_{A_i}(x)K(x, B_i).$$

  This is a product of two $\mathcal{M}$-measurable functions, hence $\mathcal{M}$-measurable. By the linearity of the integral, $T\chi_C$ is measurable for all $C \in \mathcal{A}$.
- Let $\mathcal{C} = \{C \in \mathcal{M} \otimes \mathcal{N} : T\chi_C \text{ is } \mathcal{M}\text{-measurable}\}$, then $\mathcal{C}$ is a monotone class.
  - Increasing Unions: Let $\{C_n\}_{n\in\mathbb{N}}$ be a sequence in $\mathcal{C}$ such that $C_n \uparrow C$. Then $\chi_{C_n} \uparrow \chi_C$ pointwise. By the Monotone Convergence Theorem:

    $$T\chi_C(x) = \int_{\mathcal{Y}} \chi_C(x, y)K(x, \mathrm{d}y) = \lim_{n\to\infty} \int_{\mathcal{Y}} \chi_{C_n}(x, y)K(x, \mathrm{d}y) = \lim_{n\to\infty} T\chi_{C_n}(x).$$

    Since each $T\chi_{C_n}$ is $\mathcal{M}$-measurable and the pointwise limit of measurable functions is measurable, $T\chi_C$ is measurable. Thus $C \in \mathcal{C}$.
  - Decreasing Intersections: Let $\{C_n\}_{n\in\mathbb{N}}$ be a sequence in $\mathcal{C}$ such that $C_n \downarrow C$. Then $\chi_{C_n} \downarrow \chi_C$ pointwise. Since K is a *finite* kernel, we have $T\chi_{C_n} \leq K(x, \mathcal{Y}) < \infty$. By the Dominated Convergence Theorem:

    $$T\chi_C(x) = \lim_{n\to\infty} T\chi_{C_n}(x).$$

    Again, the limit of measurable functions is measurable, so $C \in \mathcal{C}$.
- Since $\mathcal{A}$ is an algebra and $\mathcal{C}$ is a monotone class containing $\mathcal{A}$, by the Monotone Class Lemma,

  $$\mathcal{M} \otimes \mathcal{N} = \sigma(\mathcal{A}) = \mathcal{C}(\mathcal{A}) \subset \mathcal{C}.$$

  Thus, $T\chi_C$ is $\mathcal{M}$-measurable for all $C \in \mathcal{M} \otimes \mathcal{N}$.

- The result extends to simple functions by linearity and to all non-negative measurable functions f by taking a sequence $s_n \uparrow f$ and applying the Monotone Convergence Theorem, noting that the pointwise limit of measurable functions is measurable.

Finally, if K is Σ-finite, $K = \sum K_n$ for finite kernels K_n. Then $Tf = \sum T_n f$, and the sum of measurable functions remains measurable. □

Proposition 1.22. Let $(\mathcal{X}, \mathcal{M})$ and $(\mathcal{Y}, \mathcal{N})$ be measurable spaces. Let ν be a σ-finite measure on $(\mathcal{Y}, \mathcal{N})$ and let $k : \mathcal{X} \times \mathcal{Y} \to [0, \infty]$ be an $(\mathcal{M} \otimes \mathcal{N})$-measurable function. Then the mapping $K : \mathcal{X} \times \mathcal{N} \to [0, \infty]$ defined by

$$K(x, B) := \int_B k(x, y)\nu(\mathrm{d}y)$$

is a kernel from $(\mathcal{X}, \mathcal{M})$ into $(\mathcal{Y}, \mathcal{N})$.

Proof. We must verify that $K(x, \cdot)$ is a measure for each fixed x, and that $K(\cdot, B)$ is a measurable function for each fixed B.

- Fix $x \in \mathcal{X}$. The mapping $B \mapsto \int_B k(x, y)\nu(\mathrm{d}y)$ is a measure on $(\mathcal{Y}, \mathcal{N})$ by the standard construction of a measure via a density (Radon-Nikodym theorem in reverse). Specifically, $K(x, \emptyset) = 0$ and countable additivity follows from the Monotone Convergence Theorem applied to the integrand $k(x, y) \sum \chi_{B_n}(y)$.
- Fix $B \in \mathcal{N}$. We must show $x \mapsto K(x, B)$ is $\mathcal{M}$-measurable. Define the function

  $$f(x, y) = k(x, y)\chi_B(y).$$

  Since k is $(\mathcal{M} \otimes \mathcal{N})$-measurable and χ_B is trivially so, f is $(\mathcal{M} \otimes \mathcal{N})$-measurable.

  We can view the measure ν as a constant kernel $L(x, \mathrm{d}y) = \nu(\mathrm{d}y)$. Since ν is σ-finite, L is a Σ-finite kernel. Then:

  $$K(x, B) = \int_{\mathcal{Y}} f(x, y)L(x, \mathrm{d}y).$$

  By prop:kernel-measurability, this integral defines an $\mathcal{M}$-measurable function of x.

□

1.3 Radon-Nikodym Derivatives as Densities

In this section, we establish the conditions under which one measure can be represented as a density relative to another. For a more exhaustive treatment of the Lebesgue decomposition, see the notes from *MATH 507: Measure Theory and Integration.*

Definition 1.23. Let ν be a signed measure and μ a positive measure on $(\mathcal{X}, \mathcal{M})$. Then ν is said to be **absolutely continuous** w.r.t. μ, denoted $\nu \ll \mu$, if $\mu(E) = 0 \implies \nu(E) = 0$ for all $E \in \mathcal{M}$.

Theorem 1.24. *(Lebesgue-Radon-Nikodym)* Let ν be a σ-finite signed measure, and μ a positive σ-finite measure on $(\mathcal{X}, \mathcal{M})$. Then there exist unique σ-finite measures λ and ρ on $(\mathcal{X}, \mathcal{M})$ such that

$$\lambda \perp \mu, \quad \rho \ll \mu, \quad \text{and} \quad \nu = \lambda + \rho \qquad \text{(Lebesgue)}.$$

Moreover there is an extended μ-integrable function $f : \mathcal{X} \to \mathbb{R}$ such that $d\rho = f d\mu$ (i.e. such that $\rho(E) = \int_E f \, \mathrm{d}\mu$ for all $E \in \mathcal{M}$). And any two such functions are equal μ-a.e. (Radon-Nikodym). As notation we write $f = \frac{d\rho}{d\mu}$ and call this the **Radon-Nikodym derivative**.

As noted in Exercise 3.14 of Folland [1], the assumption that ν is σ-finite is not strictly necessary for the existence of a Radon-Nikodym derivative, provided we allow the density to take values in the extended real line (see prop:rn-unsigned-sigma-finite below for a precise statement).

Proposition 1.25. Let ν be an arbitrary signed measure and μ a σ-finite measure on $(\mathcal{X}, \mathcal{M})$ such that $\nu \ll \mu$. Then there exists an extended μ-integrable $f : \mathcal{X} \to [-\infty, \infty]$ such that $\mathrm{d}\nu = f \,\mathrm{d}\mu$.

The only condition required for the existence of the derivative $\frac{\mathrm{d}\nu}{\mathrm{d}\mu}$ is that the dominating measure μ be σ-finite. Indeed, if $\mathrm{d}\nu = f \,\mathrm{d}\mu$, then $\mu(A) = 0 \implies \nu(A) = \int_A f \,\mathrm{d}\mu = 0$, which shows that $\nu \ll \mu$ is both necessary and sufficient.

Proposition 1.26. Let ν and μ be measures $(\mathcal{X}, \mathcal{M})$. Suppose $\nu \ll \mu$ and μ is σ-finite. Then

$$\int f \frac{\mathrm{d}\nu}{\mathrm{d}\mu} \,\mathrm{d}\mu = \int f \,\mathrm{d}\nu$$

for all $\mathcal{M}$-measurable $f : \mathcal{X} \to [0, \infty]$.

Proof. First for any measurable E,

$$\int \chi_E \frac{d\nu}{d\mu} d\mu = \int_E \frac{d\nu}{d\mu} d\mu = \nu(E)$$

since $\frac{d\nu}{d\mu}$ is the Radon-Nikodym density (which exists by σ-finiteness of μ). On the other hand $\int \chi_E d\nu = \nu(E)$. Linearity then extends the result to simple functions. For any positive measurable $f : X \to [0, \infty]$, there is a sequence of simple functions $f_n : X \to [0, \infty)$ increasing to f. For example,

$$f_n(x) = \sum_{m=0}^{4^n-1} \frac{m}{2^n} \chi_{f^{-1}(I_{m,n})} + 2^n \chi_{f^{-1}([2^n,\infty])}, \qquad I_{m,n} = [m/2^n, (m+1)/2^n].$$

By the M.C.T. applied to $f_n \frac{d\nu}{d\mu}$, the result for simple functions, and then the M.C.T. applied to f_n:

$$\int f \frac{d\nu}{d\mu} d\mu = \int \lim_{n\to\infty} f_n \frac{d\nu}{d\mu} d\mu = \lim_{n\to\infty} \int f_n \frac{d\mu}{d\mu} d\lambda = \lim_{n\to\infty} \int f_n d\nu = \int f d\nu.$$

(Note that $\frac{d\nu}{d\mu}$ is non-negative, so $f_n \frac{d\nu}{d\mu}$ increases to $f \frac{d\nu}{d\mu}$). □

Proposition 1.27. Let μ be a σ-finite measure and $\nu \ll \mu$ with density $p = \frac{\mathrm{d}\nu}{\mathrm{d}\mu}$. Then:

(i) ν is σ-finite if and only if $p < \infty$ μ-a.e.

(ii) $\mu \ll \nu$ (and thus $\mu \equiv \nu$) if and only if $p > 0$ μ-a.e.

Proof.

(i) Suppose ν is σ-finite. Let $\bigcup_{n\in\mathbb{N}} E_n = \mathcal{X}$ be a disjoint union such that $\nu(E_n) < \infty$. Since $\nu(E_n) = \int_{E_n} p \,\mathrm{d}\mu < \infty$, the function p must be finite μ-a.e. on each E_n, and thus μ-a.e. on $\mathcal{X}$. Conversely, if $p < \infty$ μ-a.e., we can define $E_{n,m} = \{x \in F_n : m - 1 \le p(x) < m\}$ where $\{F_n\}$ is a σ-finite partition for μ. Then $\nu(E_{n,m}) = \int_{E_{n,m}} p \,\mathrm{d}\mu \le m\mu(F_n) < \infty$, providing a σ-finite partition for ν.

(ii) Suppose $\mu \ll \nu$. Let $A_0 = \{x : p(x) = 0\}$. Then $\nu(A_0) = \int_{A_0} p \,\mathrm{d}\mu = 0$. Since $\mu \ll \nu$, we must have $\mu(A_0) = 0$, so $p > 0$ μ-a.e. Conversely, if $p > 0$ μ-a.e., let $\nu(A) = 0$. Then $\int_A p \,\mathrm{d}\mu = 0$. Since p is strictly positive μ-a.e., the integrand is zero only if $\mu(A) = 0$. Thus $\mu \ll \nu$.

□

Proposition 1.28. Let ν, μ, λ be measures on $(\mathcal{X}, \mathcal{M})$. Assume all required Radon-Nikodym derivatives exist and λ is σ-finite.

(i) Linearity: For $a, b \geq 0$, $\frac{\mathrm{d}(a\nu+b\mu)}{\mathrm{d}\lambda} = a\frac{\mathrm{d}\nu}{\mathrm{d}\lambda} + b\frac{\mathrm{d}\mu}{\mathrm{d}\lambda}$ (λ-a.e.).

(ii) Chain Rule: If $\nu \ll \mu \ll \lambda$, then $\frac{\mathrm{d}\nu}{\mathrm{d}\lambda} = \frac{\mathrm{d}\nu}{\mathrm{d}\mu}\frac{\mathrm{d}\mu}{\mathrm{d}\lambda}$ (λ-a.e.).

(iii) Reciprocal: If $\nu \equiv \mu$, then $\frac{\mathrm{d}\nu}{\mathrm{d}\mu} = \left(\frac{\mathrm{d}\mu}{\mathrm{d}\nu}\right)^{-1}$ (μ-a.e. and ν-a.e.).

(iv) Quotient Rule: $\frac{\mathrm{d}\nu}{\mathrm{d}\mu} = \frac{\mathrm{d}\nu}{\mathrm{d}\lambda} / \frac{\mathrm{d}\mu}{\mathrm{d}\lambda}$ (μ-a.e.).

To prove these properties by evaluating them on sets (measures) rather than general functions, we rely on the defining property of the Radon-Nikodym derivative: p is the density $\frac{\mathrm{d}\nu}{\mathrm{d}\mu}$ if and only if $\nu(A) = \int_A p \,\mathrm{d}\mu$ for all $A \in \mathcal{M}$.

Proof. In each case, we show that the proposed density, when integrated over an arbitrary set $A \in \mathcal{M}$ with respect to the dominating measure, yields the correct measure of A.

(i) Linearity: Let $A \in \mathcal{M}$. By the definition of the measures ν and μ via their densities with respect to λ:

$$(a\nu + b\mu)(A) = a\nu(A) + b\mu(A) = a\int_A \frac{\mathrm{d}\nu}{\mathrm{d}\lambda}\,\mathrm{d}\lambda + b\int_A \frac{\mathrm{d}\mu}{\mathrm{d}\lambda}\,\mathrm{d}\lambda.$$

By the linearity of the integral:

$$= \int_A \left(a\frac{\mathrm{d}\nu}{\mathrm{d}\lambda} + b\frac{\mathrm{d}\mu}{\mathrm{d}\lambda}\right)\mathrm{d}\lambda.$$

Since this holds for all $A \in \mathcal{M}$, the integrand is the Radon-Nikodym derivative of $(a\nu + b\mu)$ with respect to λ.

(ii) Chain Rule: Let $A \in \mathcal{M}$. By the definition of the density $\frac{\mathrm{d}\nu}{\mathrm{d}\mu}$:

$$\nu(A) = \int_A \frac{\mathrm{d}\nu}{\mathrm{d}\mu}\,\mathrm{d}\mu = \int_{\mathcal{X}} \chi_A \frac{\mathrm{d}\nu}{\mathrm{d}\mu}\,\mathrm{d}\mu.$$

Since $\mu \ll \lambda$, we can apply prop:rn-integration to the function $f = \chi_A \frac{\mathrm{d}\nu}{\mathrm{d}\mu}$:

$$\int_{\mathcal{X}} \left(\chi_A \frac{\mathrm{d}\nu}{\mathrm{d}\mu}\right)\mathrm{d}\mu = \int_{\mathcal{X}} \left(\chi_A \frac{\mathrm{d}\nu}{\mathrm{d}\mu}\right)\frac{\mathrm{d}\mu}{\mathrm{d}\lambda}\,\mathrm{d}\lambda = \int_A \left(\frac{\mathrm{d}\nu}{\mathrm{d}\mu}\frac{\mathrm{d}\mu}{\mathrm{d}\lambda}\right)\mathrm{d}\lambda.$$

Thus, the product of the densities is the derivative of ν with respect to λ.

(iii) Reciprocal: Since $\nu \ll \mu$ and $\mu \ll \nu$, we can apply the Chain Rule from part (ii) with $\lambda = \nu$:

$$\frac{\mathrm{d}\nu}{\mathrm{d}\nu} = \frac{\mathrm{d}\nu}{\mathrm{d}\mu}\frac{\mathrm{d}\mu}{\mathrm{d}\nu} \quad \nu\text{-a.e.}$$

By the uniqueness of the Radon-Nikodym derivative, and noting that $\nu(A) = \int_A 1 \,\mathrm{d}\nu$ implies $\frac{\mathrm{d}\nu}{\mathrm{d}\nu} = 1$, we have $1 = \frac{\mathrm{d}\nu}{\mathrm{d}\mu}\frac{\mathrm{d}\mu}{\mathrm{d}\nu}$ ν-a.e. (and by absolute continuity, μ-a.e.). Solving for $\frac{\mathrm{d}\nu}{\mathrm{d}\mu}$ yields the reciprocal.

(iv) Quotient Rule: Applying the Chain Rule to $\nu \ll \mu \ll \lambda$ gives:

$$\frac{\mathrm{d}\nu}{\mathrm{d}\lambda} = \frac{\mathrm{d}\nu}{\mathrm{d}\mu}\frac{\mathrm{d}\mu}{\mathrm{d}\lambda}.$$

Dividing by $\frac{\mathrm{d}\mu}{\mathrm{d}\lambda}$ (which is strictly positive μ-a.e. by prop:rn-sigma-finiteness) yields the result.

□

Proposition 1.29. Suppose $\nu \ll \mu$ on $(\mathcal{X}, \mathcal{M})$. For any measurable injection $h : \mathcal{X} \to \mathcal{Y}$, the density of the pushforward measures satisfies:

$$\frac{\mathrm{d}(\nu \circ h^{-1})}{\mathrm{d}(\mu \circ h^{-1})} = \frac{\mathrm{d}\nu}{\mathrm{d}\mu} \circ h^{-1} \quad (\mu \circ h^{-1})\text{-a.e.}$$

Proof. Let f be a positive measurable function on $\mathcal{Y}$. By the definition of pushforward and Change of Variables:

$$\int_{\mathcal{Y}} f \,\mathrm{d}(\nu \circ h^{-1}) = \int_{\mathcal{X}} (f \circ h)\,\mathrm{d}\nu = \int_{\mathcal{X}} (f \circ h)\frac{\mathrm{d}\nu}{\mathrm{d}\mu}\,\mathrm{d}\mu.$$

Since h is an injection, $(f \circ h) \cdot \frac{\mathrm{d}\nu}{\mathrm{d}\mu} = (f \cdot (\frac{\mathrm{d}\nu}{\mathrm{d}\mu} \circ h^{-1})) \circ h$. Applying Change of Variables again:

$$\int_{\mathcal{X}} \left(f \cdot \left(\frac{\mathrm{d}\nu}{\mathrm{d}\mu} \circ h^{-1} \right) \right) \circ h \,\mathrm{d}\mu = \int_{\mathcal{Y}} f(y) \frac{\mathrm{d}\nu}{\mathrm{d}\mu}(h^{-1}(y))(\mu \circ h^{-1})(\mathrm{d}y).$$

The result follows from the uniqueness of the Radon-Nikodym derivative. □

Proposition 1.30. Let $(\mathcal{X}, \mathcal{M})$ and $(\mathcal{Y}, \mathcal{N})$ be measurable spaces. Let ν, μ be measures on $\mathcal{X}$ and L, K be kernels from $\mathcal{X}$ to $\mathcal{Y}$. If $p = \frac{\mathrm{d}\nu}{\mathrm{d}\mu}$ exists and $k(x, y) = \frac{\mathrm{d}L(x,\cdot)}{\mathrm{d}K(x,\cdot)}(y)$ exists as an $(\mathcal{M} \otimes \mathcal{N})$-measurable function, then the density of the marginal measure νL with respect to μK is:

$$\frac{\mathrm{d}(\nu L)}{\mathrm{d}(\mu K)}(y) = \int_{\mathcal{X}} p(x)k(x, y)M(y, \mathrm{d}x)$$

where M is the kernel obtained from the disintegration of the joint measure $\pi(\mathrm{d}x, \mathrm{d}y) = \mu(\mathrm{d}x)K(x, \mathrm{d}y)$ with respect to its marginal μK.

Proof. Let f be a positive measurable function on $\mathcal{Y}$. Using prop:kernel-measure-function:

$$(\nu L)f = \int_{\mathcal{X}} (Lf)(x)\nu(\mathrm{d}x) = \int_{\mathcal{X}} \left(\int_{\mathcal{Y}} f(y)L(x, \mathrm{d}y) \right) p(x)\mu(\mathrm{d}x).$$

Using the density $k(x, y)$, we write $L(x, \mathrm{d}y) = k(x, y)K(x, \mathrm{d}y)$:

$$= \int_{\mathcal{X}} \int_{\mathcal{Y}} f(y)k(x, y)p(x)K(x, \mathrm{d}y)\mu(\mathrm{d}x) = \int_{\mathcal{X}\times\mathcal{Y}} f(y)k(x, y)p(x)\pi(\mathrm{d}x, \mathrm{d}y).$$

By the disintegration theorem, $\pi(\mathrm{d}x, \mathrm{d}y) = (\mu K)(\mathrm{d}y)M(y, \mathrm{d}x)$. Substituting this:

$$= \int_{\mathcal{Y}} f(y) \left(\int_{\mathcal{X}} p(x)k(x, y)M(y, \mathrm{d}x) \right) (\mu K)(\mathrm{d}y).$$

Thus, the inner integral is the Radon-Nikodym derivative of νL with respect to μK. □

1.4 Joint Distributions and σ-algebras as Information

Note 1.31. Let $n \in \mathbb{N}$. For each $i \in \{1, \dots, n\}$, let $(\mathcal{X}_i, \mathcal{M}_i)$ be a measurable space. The product space is $\mathcal{X} = \prod_{i=1}^n \mathcal{X}_i$. The **product σ-algebra** $\mathcal{M} = \bigotimes_{i=1}^n \mathcal{M}_i$ is the σ-algebra on $\mathcal{X}$ generated by the algebra $\mathcal{A}$ of **measurable rectangles**:

$$\mathcal{A} = \{A_1 \times \cdots \times A_n : A_i \in \mathcal{M}_i\}.$$

The product measure on $(\mathcal{X}, \mathcal{M})$ is defined via the Carathéodory construction: it is the outer measure induced by the pre-measure on $\mathcal{A}$ (where the pre-measure of a rectangle is the product of its components' measures), restricted to $\mathcal{M}$.

Definition 1.32. Let $(\Omega, \mathcal{H}, \mathbb{P})$ be a probability space and $X_1, \dots, X_n$ be random variables where $X_i : \Omega \to \mathcal{X}_i$ is $\mathcal{M}_i$-measurable. The **joint map** $X : \Omega \to \prod \mathcal{X}_i$ is defined by:

$$X(\omega) = (X_1(\omega), \dots, X_n(\omega)).$$

The **joint distribution** of $X_1, \dots, X_n$ is the pushforward measure $\mu_X = \mathbb{P} \circ X^{-1}$ on $(\mathcal{X}, \mathcal{M})$.

Proposition 1.33. The joint map X is $\mathcal{M}$-measurable if and only if each X_i is $\mathcal{M}_i$-measurable. Furthermore, μ_X is uniquely determined by its values on measurable rectangles:

$$\mu_X(A_1 \times \cdots \times A_n) = \mathbb{P}(X_1 \in A_1, \dots, X_n \in A_n).$$

Proof. ($\Longrightarrow$) Let $\pi_i : \mathcal{X} \to \mathcal{X}_i$ be the projection $\pi_i(x) = x_i$, which is measurable by the definition of $\mathcal{M}$. Since $X_i = \pi_i \circ X$, the composition of measurable functions is measurable.
($\Longleftarrow$) For any rectangle A, $X^{-1}(A) = \bigcap_{i=1}^n X_i^{-1}(A_i)$. Since each $X_i^{-1}(A_i) \in \mathcal{H}$ and $\mathcal{H}$ is closed under finite intersection, $X^{-1}(A) \in \mathcal{H}$. Because rectangles generate $\mathcal{M}$, X is measurable. Uniqueness follows from the Carathéodory Extension Theorem, as rectangles form an algebra (and a π-system) that generates $\mathcal{M}$. $\square$

Infinite Products and Stochastic Processes

We now generalize to an arbitrary index set T (e.g., $\mathbb{N}, \mathbb{R}, [0, \infty)$) with potentially heterogeneous state spaces $(\mathcal{X}_t, \mathcal{M}_t)$ for each $t \in T$.

Definition 1.34. Let $\mathcal{X}^T = \prod_{t \in T} \mathcal{X}_t$. The **product σ-algebra** $\mathcal{M}^T = \bigotimes_{t \in T} \mathcal{M}_t$ is the σ-algebra on $\mathcal{X}^T$ generated by the collection of **cylinder sets** of the form:

$$C = \{x \in \mathcal{X}^T : x_{t_1} \in A_{t_1}, \dots, x_{t_k} \in A_{t_k}\}$$

for any finite subset $\{t_1, \dots, t_k\} \subseteq T$ and $A_{t_j} \in \mathcal{M}_{t_j}$.

Note 1.35. A **stochastic process** is the specific case where $(\mathcal{X}_t, \mathcal{M}_t) = (\mathcal{X}, \mathcal{M})$ for all $t \in T$. We call $\mathcal{X}^T$ the **path space**. The joint map $X(\omega) = (X_t(\omega))_{t \in T}$ is $\mathcal{M}^T$-measurable if and only if each coordinate X_t is $\mathcal{M}_t$-measurable. Its probability law μ_X is uniquely determined by its values on cylinder sets (finite-dimensional distributions) via Carathéodory.

Information and Generated σ-algebras

In probability theory, a sub-σ-algebra $\mathcal{G} \subset \mathcal{H}$ represents "information" or "knowledge." To know $\mathcal{G}$ is to know, for every $A \in \mathcal{G}$, whether the event A occurred.

Definition 1.36. *(Generated σ-algebras)*

1. **Single Variable:** For $X : \Omega \to \mathcal{X}$, the σ-algebra **generated by** X is the smallest sub-σ-algebra of $\mathcal{H}$ making X measurable, consisting of all pre-images:

$$\sigma(X) = X^{-1}(\mathcal{M}) = \{X^{-1}(A) : A \in \mathcal{M}\}.$$

2. **Finite Collection:** For $X_1, \dots, X_n$, the generated σ-algebra is $\sigma(X_1, \dots, X_n) = \sigma(X)$, where X is the joint map. This is the smallest σ-algebra containing all sets $\{X_i \in A_i\}$ for $i = 1, \dots, n$.

3. **Arbitrary Collection:** For a collection $\{X_t\}_{t \in T}$, the **σ-algebra generated by the collection**, denoted $\sigma(X_t : t \in T)$, is the smallest σ-algebra making every X_t measurable:

$$\sigma(X_t : t \in T) = \bigvee_{t \in T} \sigma(X_t) = \sigma\left(\bigcup_{t \in T} \sigma(X_t)\right).$$

Proposition 1.37. If $X : \Omega \to \mathcal{X}^T$ is the joint map of the collection $\{X_t\}_{t \in T}$, then:

$$\sigma(X) = X^{-1}(\mathcal{M}^T) = \sigma(X_t : t \in T).$$

Proof. By Proposition 1.33 (extended to the infinite case), X is measurable with respect to a σ-algebra $\mathcal{G}$ if and only if each X_t is measurable with respect to $\mathcal{G}$. Thus, the smallest σ-algebra satisfying one condition must satisfy the other. □

Note 1.38. *(Heuristics of Information)*

- **Indicator Functions:** If $A \in \mathcal{G}$, we "know" A because its indicator $\chi_A(\omega)$ is $\mathcal{G}$-measurable.
- **The Scale of Information:** Larger σ-algebras represent finer knowledge.
  1. **Trivial Information:** $\mathcal{G} = \{\emptyset, \Omega\}$. Only constant functions are measurable; we know nothing of the outcome.
  2. **Generated Information:** $\mathcal{G} = \sigma(X)$. We know the value $X(\omega)$, allowing us to determine $f(X(\omega))$, but we do not know ω itself.
  3. **Full Information:** $\mathcal{G} = \mathcal{H}$. We know the outcome of every possible measurable event.
- **Information Loss:** If $\sigma(Y) \subset \sigma(X)$, then X provides at least as much information as Y. For $X \sim \mathcal{N}(0, 1)$ and $Y = |X|$, $\sigma(Y) \subsetneq \sigma(X)$. Knowing X determines Y, but knowing Y leaves the sign of X uncertain.

Proposition 1.39. *(Doob-Dynkin Lemma)* Let $V : \Omega \to \mathbb{R}$ be a random variable.

(i) **Finite Case:** V is $\sigma(X_1, \dots, X_n)$-measurable if and only if there exists a measurable function $f : \mathcal{X}_1 \times \cdots \times \mathcal{X}_n \to \mathbb{R}$ such that $V = f(X_1, \dots, X_n)$.

(ii) **Infinite Case:** Let T be an arbitrary index set. V is $\sigma(X_t : t \in T)$-measurable if and only if there exists a **countable** sequence of indices $\{t_1, t_2, \dots\} \subseteq T$ and a measurable function $f : \prod_{j=1}^{\infty} \mathcal{X}_{t_j} \to \mathbb{R}$ such that:

$$V = f(X_{t_1}, X_{t_2}, \dots).$$

Proof. Omitted. (Note: The infinite case highlights that even in uncountable processes, any single $\sigma(X_t : t \in T)$-measurable random variable only depends on at most countably many coordinates). □

1.5 Modes of Convergence

We now study the various ways in which a sequence of random variables $\{X_n\}_{n\in\mathbb{N}}$ can converge to a limit X. Throughout, let $(\Omega, \mathcal{H}, \mathbb{P})$ be a probability space and $X, \{X_n\}_{n\in\mathbb{N}}$ be real-valued random variables. Many proofs appeal to concentration inequalities which I have irresponsibly postponed.

Definition 1.40. We define the following two primary modes of convergence:

(i) X_n **converges almost surely** to X, denoted $X_n \xrightarrow{a.s.} X$, if

$$\mathbb{P}\left(\left\{\omega \in \Omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\right\}\right) = 1.$$

(ii) X_n **converges in probability** to X, denoted $X_n \xrightarrow{P} X$, if for every $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}(|X_n - X| > \varepsilon) = 0.$$

Proposition 1.41. *(Continuous Mapping Theorem for a.s. Convergence)* Let $f : \mathbb{R}^K \to \mathbb{R}$ be a function. If $X_{k,n} \xrightarrow{a.s.} X_k$ for each $k = 1, \ldots, K$, and f is continuous on a set $A \subset \mathbb{R}^K$ such that $\mathbb{P}((X_1, \ldots, X_K) \in A) = 1$, then

$$f(X_{1,n}, \ldots, X_{K,n}) \xrightarrow{a.s.} f(X_1, \ldots, X_K).$$

Proof. Let $\Omega_k = \{\omega \in \Omega : \lim X_{k,n}(\omega) = X_k(\omega)\}$ and $\Omega_A = \{\omega \in \Omega : (X_1(\omega), \ldots, X_K(\omega)) \in A\}$. By assumption, $\mathbb{P}(\Omega_k) = 1$ and $\mathbb{P}(\Omega_A) = 1$. Let $\Omega_0 = \Omega_A \cap \left(\bigcap_{k=1}^K \Omega_k\right)$. Then $\mathbb{P}(\Omega_0) = 1$. For any $\omega \in \Omega_0$, the sequence of vectors $(X_{1,n}(\omega), \ldots, X_{K,n}(\omega))$ converges to $(X_1(\omega), \ldots, X_K(\omega)) \in A$. And by the continuity of f on A, it follows that $f(X_{1,n}(\omega), \ldots, X_{K,n}(\omega)) \to f(X_1(\omega), \ldots, X_K(\omega))$. □

Proposition 1.42. $X_n \xrightarrow{a.s.} X$ if and only if

$$\sum_{n\in\mathbb{N}} \chi_{(\varepsilon,\infty)} \circ |X_n - X| < \infty \qquad a.s.$$

for all $\varepsilon > 0$. Moreover, the following are equivalent:

(i) X_n converges almost surely

(ii) $\sup_{i,j\geq n} |X_i - X_j| \xrightarrow{a.s.} 0$

(iii) $\sup_k |X_{n+k} - X_n| \xrightarrow{a.s.} 0$

Proof. Omitted. □

Lemma 1.43. *(Borel-Cantelli)* Let $\{H_n\}_{n\in\mathbb{N}}$ be a sequence of events. If $\sum_{n=1}^\infty \mathbb{P}(H_n) < \infty$, then

$$\mathbb{P}(\limsup H_n) = 0, \quad \text{or equivalently, } \sum_{n=1}^\infty \chi_{H_n} < \infty \text{ a.s.}$$

Proof. By the Monotone Convergence Theorem, $\mathbb{E}[\sum \chi_{H_n}] = \sum \mathbb{E}[\chi_{H_n}] = \sum \mathbb{P}(H_n)$. Since the sum of probabilities is finite, the expectation of the random sum is finite. A non-negative random variable with finite expectation must be finite almost surely. □

Proposition 1.44. A sequence of random variables X_n converges almost surely to X if either of the following conditions holds:

(i) For every $\varepsilon > 0$, $\sum_{n=1}^{\infty} \mathbb{P}(|X_n - X| > \varepsilon) < \infty$.

(ii) There exists a sequence $\varepsilon_n > 0$ such that $\sum_{n=1}^{\infty} \varepsilon_n < \infty$ and $\sum_{n=1}^{\infty} \mathbb{P}(|X_{n+1} - X_n| > \varepsilon_n) < \infty$.

Proof.

(i) For a fixed $\varepsilon > 0$, let $H_n = \{|X_n - X| > \varepsilon\}$. By the Borel-Cantelli Lemma, $\sum \mathbb{P}(H_n) < \infty$ implies that $\mathbb{P}(\limsup H_n) = 0$. Thus, for almost all ω, $|X_n(\omega) - X(\omega)| > \varepsilon$ for only finitely many n. Since this holds for all $\varepsilon > 0$, $X_n \xrightarrow{a.s.} X$ follows by prop:as-convergence-characterization.

(ii) Let $H_n = \{|X_{n+1} - X_n| > \varepsilon_n\}$. By Borel-Cantelli, $|X_{n+1} - X_n| \leq \varepsilon_n$ for all $n \geq N(\omega)$ almost surely. Since $\sum \varepsilon_n < \infty$, the sequence is almost surely Cauchy:

$$|X_{n+k} - X_n| \leq \sum_{j=n}^{n+k-1} |X_{j+1} - X_j| \leq \sum_{j=n}^{\infty} \varepsilon_j \xrightarrow{n\to\infty} 0.$$

By the completeness of $\mathbb{R}$ and prop:as-convergence-characterization, X_n converges almost surely.

□

Proposition 1.45. $X_n \xrightarrow{P} X$ if and only if $\mathbb{E}[\min\{1, |X_n - X|\}] \to 0$.

Proof. ($\Longrightarrow$) Suppose $X_n \xrightarrow{P} X$. Fix $\varepsilon > 0$.

$$\begin{aligned}\mathbb{E}[\min\{1, |X_n - X|\}] &= \mathbb{E}[(\min\{1, |X_n - X|\})\chi_{\{|X_n - X| > \varepsilon\}}] + \mathbb{E}[(\min\{1, |X_n - X|\})\chi_{\{|X_n - X| \leq \varepsilon\}}] \\ &\leq 1 \cdot \mathbb{P}(|X_n - X| > \varepsilon) + \varepsilon \cdot \mathbb{P}(|X_n - X| \leq \varepsilon).\end{aligned}$$

Taking $n \to \infty$, the first term goes to 0 by convergence in probability. Since ε is arbitrary, the expectation goes to 0.
($\Longleftarrow$) By Markov's Inequality, for any $\varepsilon \in (0, 1)$:

$$\mathbb{P}(|X_n - X| > \varepsilon) = \mathbb{P}(\min\{1, |X_n - X|\} > \varepsilon) \leq \frac{\mathbb{E}[\min\{1, |X_n - X|\}]}{\varepsilon}.$$

If the expectation tends to 0, then $\mathbb{P}(|X_n - X| > \varepsilon) \to 0$ for all ε. □

Theorem 1.46. The following relations hold between the modes of convergence:

(i) If $X_n \xrightarrow{a.s.} X$, then $X_n \xrightarrow{P} X$.

(ii) If $X_n \xrightarrow{P} X$, then there exists a subsequence $\{X_{n_k}\}$ such that $X_{n_k} \xrightarrow{a.s.} X$.

(iii) If every subsequence of $\{X_n\}$ has a further subsequence converging to X a.s., then $X_n \xrightarrow{P} X$.

Proof. Omitted. □

Proposition 1.47. If $X_n \xrightarrow{P} X$ and f is continuous, then $f(X_n) \xrightarrow{P} f(X)$.

Proof. Let $\{n_i\}$ be an arbitrary subsequence of indices. Since $X_n \xrightarrow{P} X$, the subsequence $\{X_{n_i}\}$ also converges in probability to X. By thm:convergence-relations(ii), there exists a further sub-subsequence $\{n_{i_j}\}$ such that $X_{n_{i_j}} \xrightarrow{a.s.} X$. Because f is continuous, the Continuous Mapping Theorem for almost sure convergence (prop:cmt-as) implies that $f(X_{n_{i_j}}) \xrightarrow{a.s.} f(X)$. Thus, every subsequence of $\{f(X_n)\}$ contains a further sub-subsequence that converges almost surely to $f(X)$. By the characterization in thm:convergence-relations(iii), it follows that $f(X_n) \xrightarrow{P} f(X)$. □

While almost sure convergence and convergence in probability describe the behavior of the random variables as functions, L^p convergence describes the behavior of their moments.

Definition 1.48. For $p \in [1, \infty)$, we say that a sequence of random variables $\{X_n\}$ **converges in** L^p to X, denoted $X_n \xrightarrow{L^p} X$, if $X_n, X \in L^p$ (i.e., $\mathbb{E}|X_n|^p < \infty$) and $\lim_{n\to\infty} \mathbb{E}[|X_n - X|^p] = 0$.

Proposition 1.49. $X_n \xrightarrow{L^p} X \implies X_n \xrightarrow{P} X$.

Proof. This follows immediately from Markov's inequality: $\mathbb{P}(|X_n - X| > \varepsilon) \leq \varepsilon^{-p}\mathbb{E}|X_n - X|^p$. □

Definition 1.50. A collection of random variables $\{X_t : t \in T\}$ is **uniformly integrable** (UI) if

$$\lim_{b\to\infty} \sup_{t\in T} \mathbb{E}\left[\chi_{\{|X_t|>b\}}|X_t|\right] = 0.$$

Theorem 1.51. *(Vitali Convergence Theorem)* Let $X_n, X \in L^1$. Then $X_n \xrightarrow{L^1} X$ if and only if $X_n \xrightarrow{P} X$ and $\{X_n\}$ is uniformly integrable.

Proof. Omitted. □

Definition 1.52. Let μ, $\{\mu_n\}_{n\in\mathbb{N}}$ be probability measures on $(\mathcal{X}, \mathcal{M})$. We say that μ_n **converges weakly** to μ, denoted $\mu_n \xrightarrow{w} \mu$, if for every bounded, continuous function $f : \mathcal{X} \to \mathbb{R}$:

$$\int f \, \mathrm{d}\mu_n \to \int f \, \mathrm{d}\mu.$$

If $X_n \sim \mu_n$ and $X \sim \mu$, we say X_n **converges in distribution** to X, denoted $X_n \xrightarrow{d} X$.

Note 1.53. The definition of weak convergence bridges probability theory and functional analysis.

1. **The Dual Pairing:** Let $V = C_b(\mathcal{X})$ be the Banach space of bounded continuous functions on $\mathcal{X}$ equipped with the supremum norm $\|f\|_\infty = \sup_{x\in\mathcal{X}} |f(x)|$. We define, for each measure μ, an associated functional $T : V \to \mathbb{R}$ via the integral:

$$T_\mu f := \int_{\mathcal{X}} f(x)\mu(\mathrm{d}x) = \mathbb{E}[f(X)].$$

   In this context, the measure μ acts as a continuous linear functional on the space of functions.

2. **Radon Measures:** A Borel measure μ is called a **Radon measure** if it is *inner regular*, meaning the measure of any Borel set can be approximated from within by compact sets: $\mu(B) = \sup\{\mu(K) : K \subset B, K \text{ compact}\}$. On most spaces of interest, every probability measure is a Radon measure.

3. **Weak vs. Weak* Convergence:** The *Riesz-Markov-Kakutani Representation Theorem* identifies the dual space V^* of $C_b(\mathcal{X})$ as the space of finite signed Radon measures. In functional analysis we have the following definitions:

   - **Weak convergence** of $\mu_n \to \mu$ would require $\Lambda(\mu_n) \to \Lambda(\mu)$ for all $\Lambda \in V^{**}$.
   - **Weak* convergence** of $\mu_n \to \mu$ requires $T_{\mu_n} f \to T_\mu f$ for all $f \in V$.

   Thus, what probabilists call "weak convergence" is technically weak* convergence on the space of measures. The term "weak" is used in probability I think because convergence in distribution is the least restrictive mode in the hierarchy of convergence.

Proposition 1.54. $X_n \xrightarrow{P} X \implies X_n \xrightarrow{d} X$.

Proof. Let f be bounded and continuous. Since $X_n \xrightarrow{P} X$, by prop:cmt-prob, $f(X_n) \xrightarrow{P} f(X)$. Since f is bounded, the Dominated Convergence Theorem implies $\mathbb{E}[f(X_n)] \to \mathbb{E}[f(X)]$. □

Theorem 1.55. *(Portmanteau)* The following are equivalent for probability measures μ_n, μ:

(i) $\mu_n \xrightarrow{w} \mu$.

(ii) $\limsup \mu_n(F) \leq \mu(F)$ for every closed set $F \subset \mathcal{X}$.

(iii) $\liminf \mu_n(G) \geq \mu(G)$ for every open set $G \subset \mathcal{X}$.

(iv) $\lim \mu_n(A) = \mu(A)$ for every Borel set A such that $\mu(\partial A) = 0$ (the μ-continuity sets).

Proof. Omitted. □

Proposition 1.56. *(Uniqueness of Weak Limits)* If $\mu_n \xrightarrow{w} \mu$ and $\mu_n \xrightarrow{w} \nu$, then $\mu = \nu$.

Proof. Firstly, $\int f \, \mathrm{d}\mu = \int f \, \mathrm{d}\nu$ for all bounded continuous f. Taking $\mu_n = \mu$ as a constant sequence and applying Portmanteau, we have $\mu(G) \leq \nu(G)$ for all open G. By symmetry, $\mu(G) = \nu(G)$ for all open sets. Thus, by outer-regularity $\mu(A) = \nu(A)$ for all $A \in \mathcal{B}(\mathbb{R})$. □

Theorem 1.57. *(Skorokhod Representation Theorem)* Let $X_n \xrightarrow{d} X$. There exists a probability space $(\Omega', \mathcal{H}', \mathbb{P}')$ and random variables Y_n, Y on it such that $Y_n \overset{d}{=} X_n$, $Y \overset{d}{=} X$, and $Y_n \xrightarrow{a.s.} Y$.

Proof. Omitted. □

Proposition 1.58. *(Continuous Mapping Theorem)* Let $h : \mathcal{X} \to \mathcal{Y}$ be an a.s. continuous measurable function. If $X_n \xrightarrow{d} X$, then $h(X_n) \xrightarrow{d} h(X)$.

Proof. There exists Skorokhod representations $\{Y_n\}_{n\in\mathbb{N}}$ and Y such that $Y_n \overset{d}{=} X_n$, $Y \overset{d}{=} X$, and $Y_n \xrightarrow{a.s.} Y$. Then by prop:cmt-as, $f(Y_n) \xrightarrow{a.s.} f(Y)$. Finally, one can show that $f(Y_n) \overset{d}{=} f(X_n)$ and $f(Y) \overset{d}{=} f(X)$. So another use of the Skorokhod representation theorem yields $f(X_n) \xrightarrow{d} f(X)$. □

Theorem 1.59. *(Lévy Continuity Theorem)* Let $\phi_n(r) = \mathbb{E}[e^{irX_n}]$ be the characteristic functions. Then $X_n \xrightarrow{d} X$ if and only if $\phi_n(r) \to \phi(r)$ pointwise, where ϕ is continuous at 0.

Proof. Omitted. □

When dealing with sequences of random vectors $\{(X_{1,n}, \dots, X_{K,n})\}_{n\in\mathbb{N}}$ in $\mathbb{R}^K$, we distinguish between the convergence of individual coordinates (**marginal convergence**) and the convergence of the vector (**joint convergence**).

Definition 1.60. We say $X_n = (X_{1,n}, \dots, X_{K,n})$ **converges jointly** to $X = (X_1, \dots, X_K)$ in mode $\alpha \in \{a.s., P, L^p, d\}$ if the sequence converges in the product space $\mathbb{R}^K$ with respect to the Borel σ-algebra $\mathcal{B}(\mathbb{R}^K)$.

Proposition 1.61. Let $\alpha \in \{a.s., P, L^p\}$. Then $X_{k,n} \xrightarrow{\alpha} X_k$ for each $k = 1, \dots, K$ if and only if $(X_{1,n}, \dots, X_{K,n}) \xrightarrow{\alpha} (X_1, \dots, X_K)$.

Proof. ($\Longleftarrow$) follows from the Continuous Mapping Theorem using the coordinate projection

$$f_k(x_1, \dots, x_k) = x_k.$$

($\Longrightarrow$) follows from the fact that the product metric $d(x, y) = (\sum |x_k - y_k|^2)^{1/2}$ is bounded by the sum of the coordinate distances. For instance, for $X_n \xrightarrow{P} X$, we use:

$$\mathbb{P}\left(\sqrt{\sum |X_{k,n} - X_k|^2} > \varepsilon\right) \leq \mathbb{P}\left(\sum |X_{k,n} - X_k| > \varepsilon\right) \leq \sum_{k=1}^{K} \mathbb{P}\left(|X_{k,n} - X_k| > \frac{\varepsilon}{K}\right).$$

Since each term in the finite sum vanishes as $n \to \infty$, the joint probability vanishes. □

Note 1.62. Marginal convergence in distribution does **not** imply joint convergence in distribution:

$$X_{k,n} \xrightarrow{d} X_k \quad \not\Longrightarrow \quad (X_{1,n}, \dots, X_{K,n}) \xrightarrow{d} (X_1, \dots, X_K).$$

Theorem 1.63. *(Cramér-Wold Device)* A sequence of random vectors $X_n \in \mathbb{R}^K$ converges in distribution to X if and only if every linear combination of their components converges in distribution:

$$X_n \xrightarrow{d} X \iff t^\top X_n \xrightarrow{d} t^\top X \quad \forall t \in \mathbb{R}^K.$$

Proof. Omitted. □

Lemma 1.64. *(Slutsky's Theorem)* Let X_n, X and Y_n be random variables. If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{P} c$ for some constant $c \in \mathbb{R}$, then the joint convergence is "rescued": $(X_n, Y_n) \xrightarrow{d} (X, c)$. Consequently:

(i) $X_n + Y_n \xrightarrow{d} X + c$

(ii) $X_n Y_n \xrightarrow{d} cX$

(iii) $X_n / Y_n \xrightarrow{d} X/c$, provided $c \neq 0$.

Proof. First, one shows $(X_n, Y_n) \xrightarrow{d} (X, c)$. This is usually done by showing the joint characteristic function $\mathbb{E}[e^{i(r_1 X_n + r_2 Y_n)}]$ converges to $\mathbb{E}[e^{ir_1 X}]e^{ir_2 c}$. Once joint convergence is established, (i)-(iii) follow by applying the Continuous Mapping Theorem to the functions $g(x, y) = x + y$, xy, and x/y respectively. □

2 Conditioning and Independence

2.1 Independence

2.2 Conditional Expectations

2.3 Conditional Distributions

2.4 Conditional Independence

3 Concentration Inequalities and Divergences

3.1 Concentration Inequalities

Proposition 3.1. *(Markov's Inequality)* Let X be as positive real-valued random variable, then

$$\mathbb{P}(X \geq t) \leq t^{-1}\mathbb{E}X$$

for any $t > 0$.

Proof. Notice $t^{-1}X(\omega) \geq \chi_{[t,\infty)}(X(\omega))$ for any positive random variable X and $t > 0$. Hence

$$t^{-1}\mathbb{E}X \geq \mathbb{E}\left[\chi_{[t,\infty)}(X(\omega))\right] = \int \chi_{[t,\infty)} \, \mathrm{d}\big(\mathbb{P} \circ X^{-1}\big) = \big(\mathbb{P} \circ X^{-1}\big)([t,\infty)) = \mathbb{P}(X \geq t).$$

□

Proposition 3.2. *(Generalized Markov's Inequality)* Let $f : \mathbb{R} \to \mathbb{R}$ be a monotone non-decreasing, positive function, X be a real-valued random variable, and $t \in \mathbb{R}$. Then

$$\mathbb{P}(X \geq t) \leq f(t)^{-1}\mathbb{E}\left[f(X)\right].$$

Proof. Since f is monotone non-decreasing, $\{\omega \in \Omega : X(\omega) \geq t\} \subset \{\omega \in \Omega : f(X(\omega)) \geq f(t)\}$. So

$$\mathbb{P}(X \geq t) \leq \mathbb{P}(f(X) \geq f(t)).$$

Since the range of f is positive and since $f(X)$ is itself a random variable, we apply Markov's Inequality to obtain

$$\mathbb{P}(X \geq t) \leq \mathbb{P}(f(X) \geq f(t)) \leq f(t)^{-1}\mathbb{E}\left[f(X)\right].$$

□

3.2 Integral Probability Metrics

3.3 ϕ–divergences

4 Martingales

References

[1] Gerald B Folland. *Real Analysis: Modern Techniques and Their Applications*, volume 168. John Wiley & Sons, New York, 2nd edition, 1999.